

# AI Methodology Report

Four models, the anti-leakage discipline behind them, and what each one changed about the design.

06

OF 12 UPLOAD SLOTS

**R<sup>2</sup> 0.9944**

SHADE SURROGATE, TEST SET

**97.5%**

COMFORT BAND ACCURACY

**k = 2**

MICROCLIMATE REGIMES

**4,402**

DAYLIGHT HOURS MODELLED

▶ VISIT THE LIVE PROJECT PORTAL

[wasim437.github.io/AI\\_SAFA/](https://wasim437.github.io/AI_SAFA/)

01

## WHAT IS IN THIS FILE

Phase9 AI Workflow and Visualization Report

DRAWING

Fig05 surrogate performance

FIGURE

Fig06 feature importance

FIGURE

Fig07 confusion matrix

FIGURE

02

## THE SCALE OF THE ANALYSIS BEHIND IT

**39 years**

of climate normals (NCM, 1977–2015) rebuilt into an 8,760-hour modelled year

**8,760 hours**

of sun position ray-traced against 131 trees for every hour of the year

**15,000**

one-metre ground cells modelled for shade and comfort, site-wide

**4 models**

trained and tested for leakage — two of them decide what this document reports

**41 checks**

run on every rebuild, so a wrong number fails loudly instead of shipping quietly

03

## THE PHASES BEHIND THIS FILE

P9 AI Workflow & Visualisation

### VERIFY THIS DOCUMENT

Every quantity in this file is regenerated from data by code. Nothing is typed by hand.

Repository [github.com/wasim437/AI\\_SAFA](https://github.com/wasim437/AI_SAFA)

Live portal [wasim437.github.io/AI\\_SAFA/](https://wasim437.github.io/AI_SAFA/)

Produced by **CODE** [src/models.py](#) [src/dataset.py](#) **MODELS** [models/model\\_metrics.json](#) **TESTS** [tests/test\\_pipeline.py](#)

Reproduce `python run_analysis.py` · `python -m tests.test_pipeline`

# A ten-phase process, checked at every step

Nothing downstream is hand-drawn or hand-typed — change an input and every figure moves with it, or a test fails loudly.

06

OF 12

## 01 THE TEN PHASES

### P1 Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

### P3 Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

### P5 Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m<sup>2</sup>

### P7 Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

### P9 AI Workflow & Visualisation

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

THIS DOCUMENT

### P2 Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

### P4 Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

### P6 Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

### P8 User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

### P10 Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

## 02 THE CODE AND DATA BEHIND THIS DOCUMENT

### Models

src

### Dataset

src

### Model metrics

models

### Test pipeline

tests

## 03 REPRODUCE IT END TO END

```
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m src.drawings           # section, elevation, planting
python -m src.boards             # the presentation boards
python tools/build_submission_pdfs.py
python -m tests.test_pipeline    # 41 checks
```

UPLOAD SLOT 06 OF 12 · AI METHODOLOGY REPORT

# AI Methodology Report

Four models, the discipline behind them, and what each one changed about the design

The challenge asks how AI supported the design. This submission's answer is a reproducible analysis pipeline that runs end to end with one command — not a description of prompting an image generator. Analysis that does not change a drawing is decoration; each model below is reported with the design decision it altered.

Al Safa 2 Park · **Falaj Al Safa** · 15,000 m<sup>2</sup> · Al Safa 2, Dubai · Mohamed Wasim, Individual Applicant · Issued 03 August 2026 · every quantity regenerated by `python run_analysis.py`

## 1. The discipline that makes these models mean anything

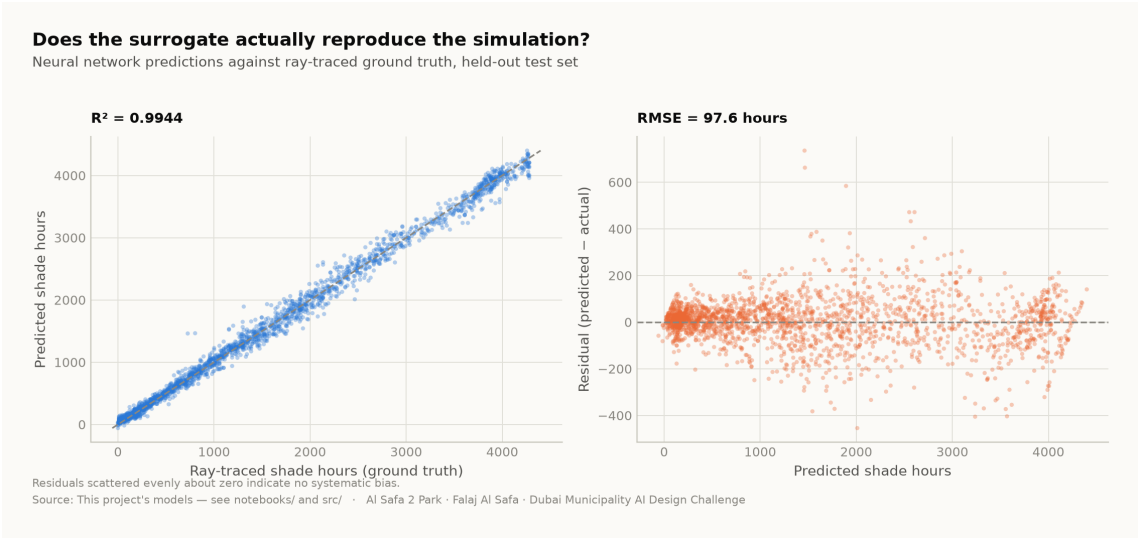
**The target must not be recoverable from the inputs by algebra.** It is easy to score  $R^2 = 1.000$  on this project by accident: the heat index is a closed-form function of temperature and humidity, so 'predicting' it from temperature and humidity is arithmetic wearing a lab coat. Every model below was designed against that failure, and the pipeline asserts the absence of leaked features as a test.

## 2. The four models

| Model                       | Task                         | Result      | Why it is a real problem                                                              |
|-----------------------------|------------------------------|-------------|---------------------------------------------------------------------------------------|
| <b>M1a</b> Random Forest    | Shade surrogate (regression) | $R^2$ 0.998 | Learns a slow ray-traced simulation from cheap plan geometry                          |
| <b>M1b</b> Neural network   | Shade surrogate (deployed)   | $R^2$ 0.994 | Differentiable, so it can sit inside a layout optimiser; a forest cannot              |
| <b>M2</b> Gradient Boosting | Comfort band (4-class)       | 97.5%       | Temperature and humidity <b>withheld</b> — it sees only sun position and the calendar |
| <b>M3</b> K-Means           | Microclimate regimes         | $k=2$       | Unsupervised; $k$ is <i>selected</i> by silhouette score, not chosen to look tidy     |

## 3. Why M1 is the flagship

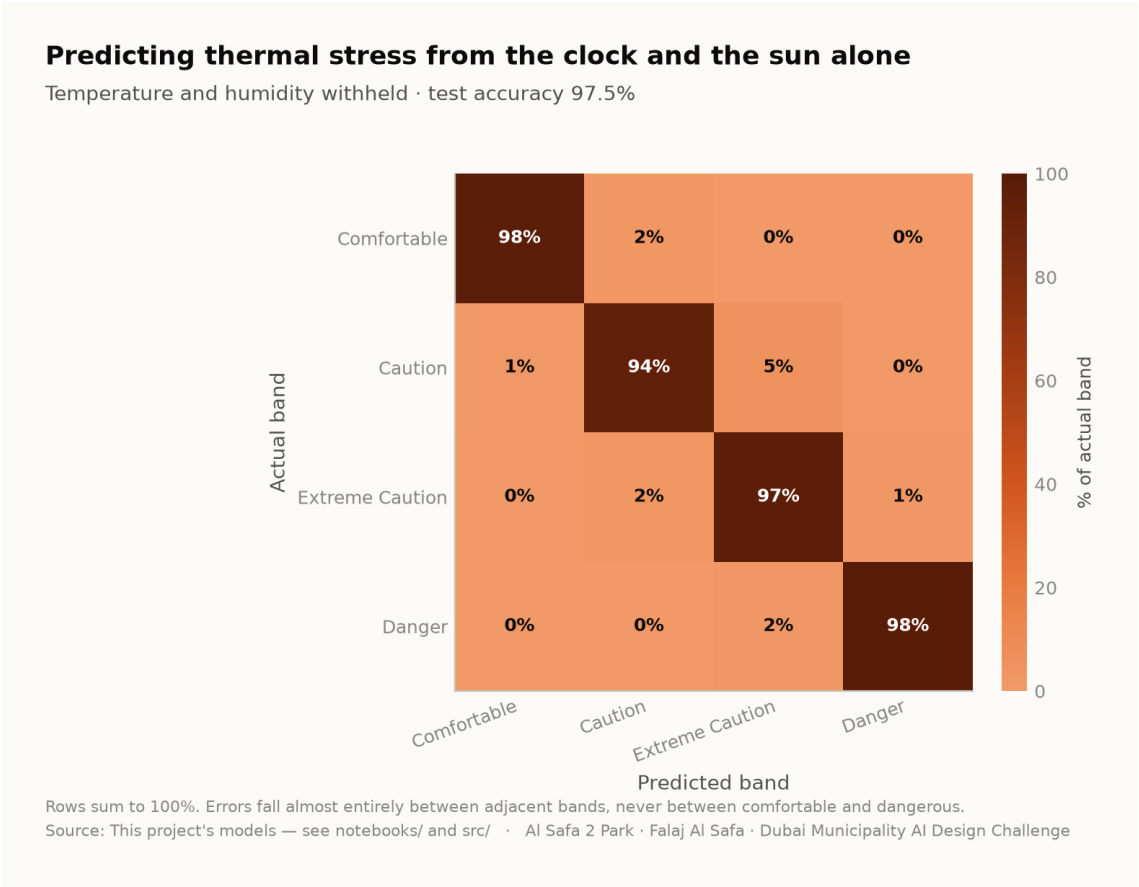
Ray-tracing annual shade means projecting 131 tree shadows onto 15,000 ground cells for every daylight hour. It is slow, and it must be re-run for every design variation. That cost is precisely what stops a designer exploring. The surrogate answers in milliseconds instead of minutes, at  $R^2$  0.994 against ray-traced ground truth on a held-out test set. That is AI changing how the design was made, rather than decorating it afterwards.



Neural-network predictions against ray-traced ground truth on the held-out test set. Analysis output — computed from project data.

4. Why M2 answers a question worth asking

If thermal stress is predictable from a clock and an ephemeris alone, park operations can be scheduled without a sensor network — a real capital and maintenance saving over the life of the park. At 97.5% accuracy with temperature and humidity withheld, it is. Errors fall between adjacent comfort bands, never between comfortable and dangerous.



Confusion matrix, temperature and humidity withheld. Analysis output — computed from project data.

5. What the models actually changed

| Model output                                                                    | Design consequence                                                                                                                     |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| A straight route has no shade anywhere for 330 hours a year; an 18 m arc has 52 | The plan is <b>an arc</b> , and the sagitta is the value that minimises that column — not the one that maximises mean cover            |
| A closed loop scores worst on every measure                                     | The circuit is kept but <b>unshaded</b> — Al Madar is a running loop, not a second canopy                                              |
| The concave side is measurably cooler than the convex side                      | The basin, quiet garden, play area and picnic grove are all placed in the hollow; the sports lawn, plaza and souk take the convex face |
| Comfort is ~97% predictable from sun and calendar                               | <b>No sensor network.</b> Programming runs off an almanac                                                                              |
| Comfort gain concentrates in late afternoon, spring and autumn                  | Activation targets those windows; summer midday is not programmed outdoors                                                             |
| An open water channel in Dubai evaporates                                       | The falaj is set on the canopy's <b>drip line</b> , shaded all day, rather than in the open                                            |

## 6. Where AI was deliberately not used

**Visitor demand is a scenario model, not a machine learning result.** No footfall data exists for this site. A model trained on invented demand would only recover the assumption that produced it, and reporting that as a finding would be dishonest. It is therefore excluded from the model suite and labelled a scenario wherever it appears.

**The photoreal renders are illustrations, not analysis.** They are captioned as artistic impressions throughout. Three earlier visuals that presented invented data as measurement were withdrawn entirely.

## 7. Reproducibility

The complete analysis — data, code, trained models and tests — runs end to end with `python run_analysis.py` and is verified by `python -m tests.test_pipeline` (38 checks), `node docs/_PORTAL/selftest.js` (64 checks) and a frame-by-frame test of the concept film. A juror who wants to know where a number comes from can be given a file path rather than an opinion.

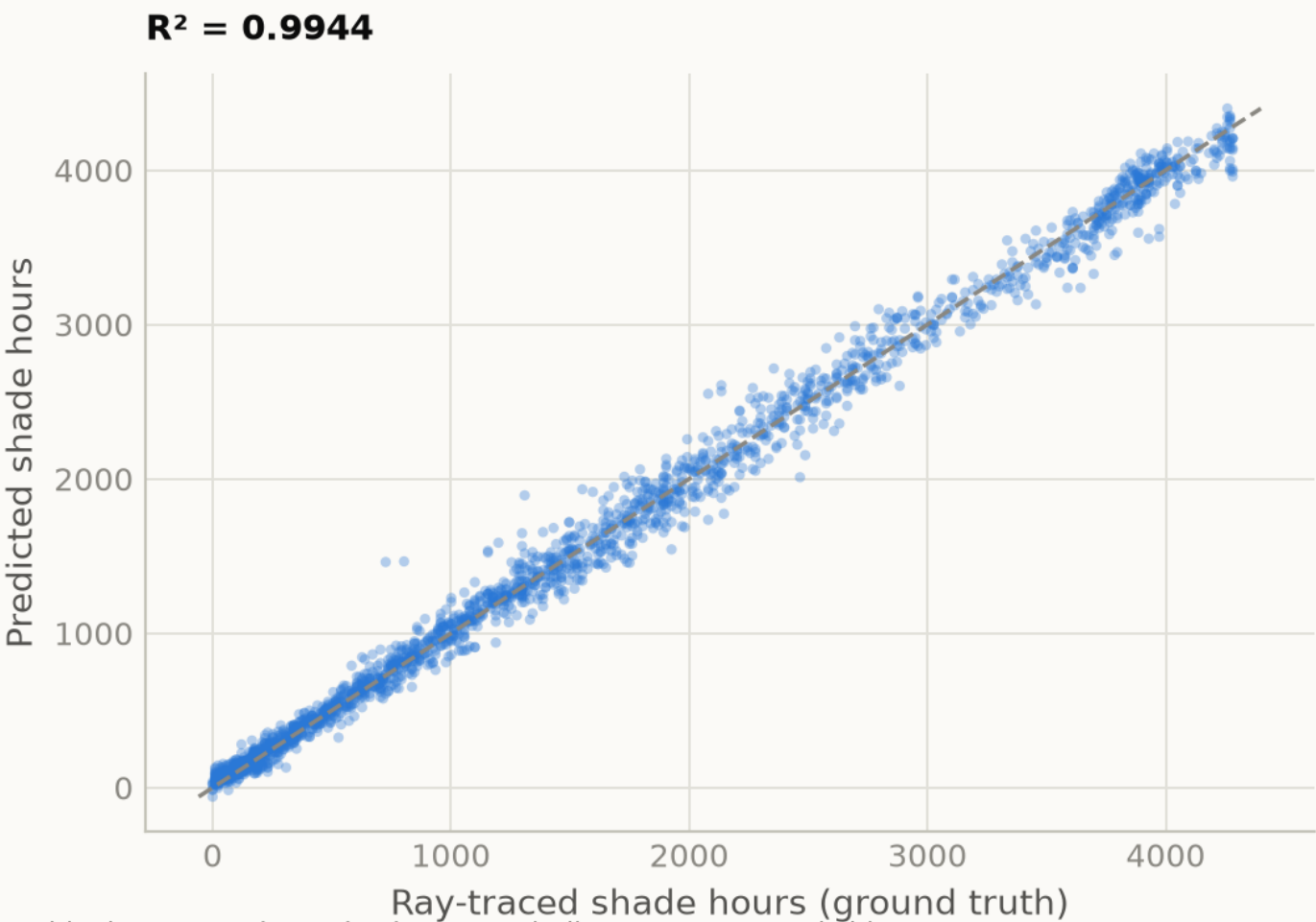
The full repository — every dataset, every model, every line of the pipeline above — is published at [https://github.com/wasim437/Al\\_SAFA](https://github.com/wasim437/Al_SAFA) and the live analytics portal at [https://wasim437.github.io/Al\\_SAFA/](https://wasim437.github.io/Al_SAFA/). Both run from the same code that produced this document.

Fig05 surrogate performance

Analysis output — computed from project data.

Does the surrogate actually reproduce the simulation?

Neural network predictions against ray-traced ground truth, held-out test set



Residuals scattered evenly about zero indicate no systematic bias.

Source: This project's models — see notebooks/ and src/ · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

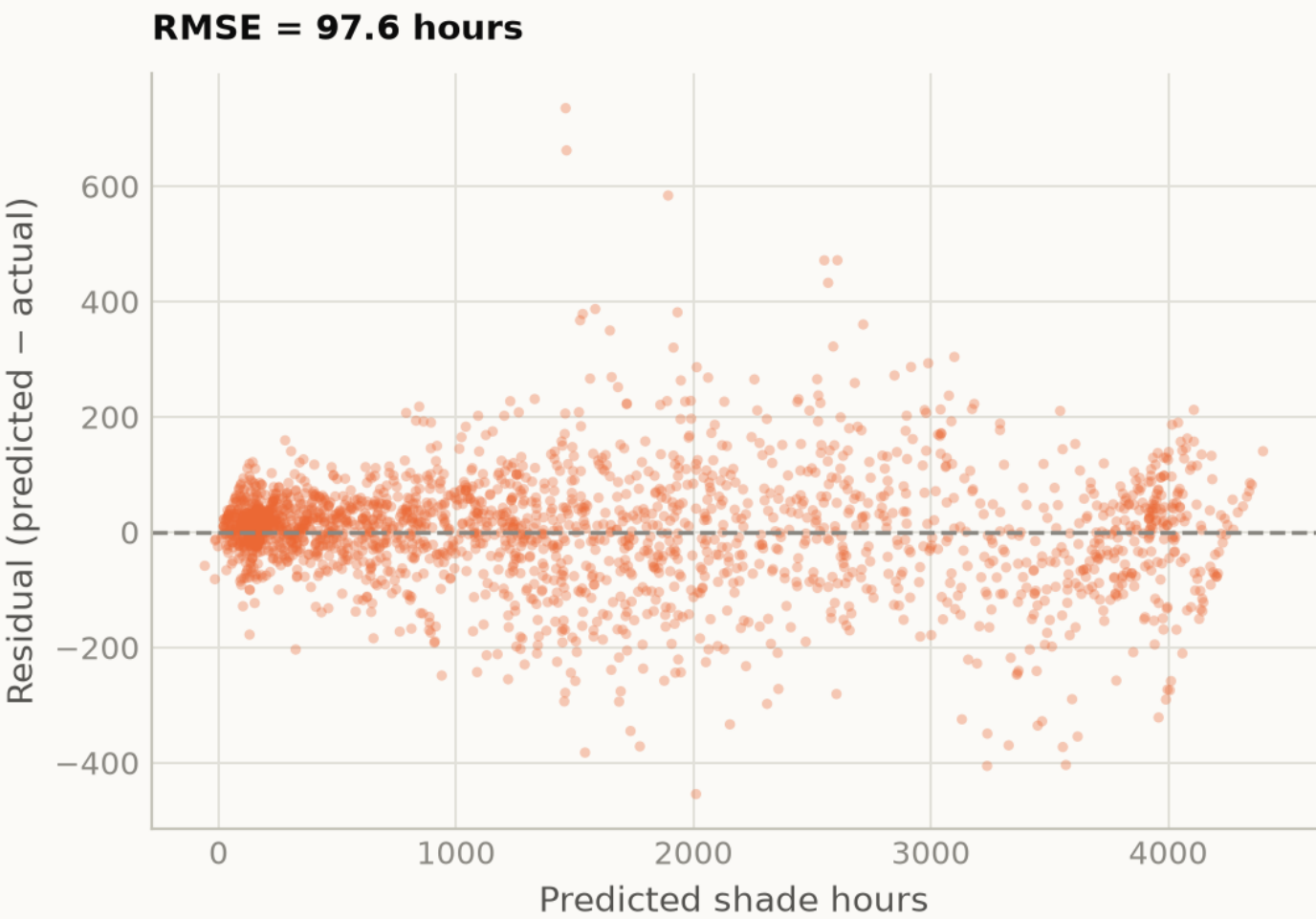
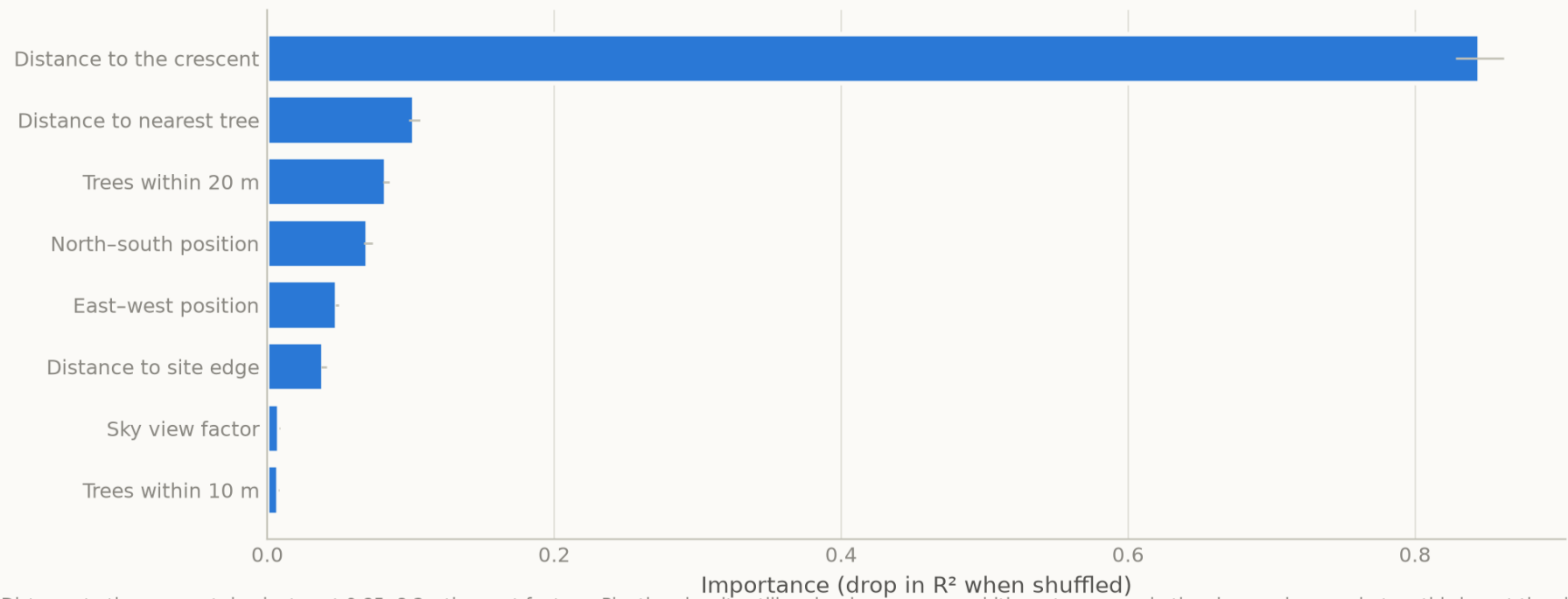


Fig06 feature importance

Analysis output — computed from project data.

What actually determines whether a square metre is shaded

Permutation importance — the drop in accuracy when each feature is shuffled



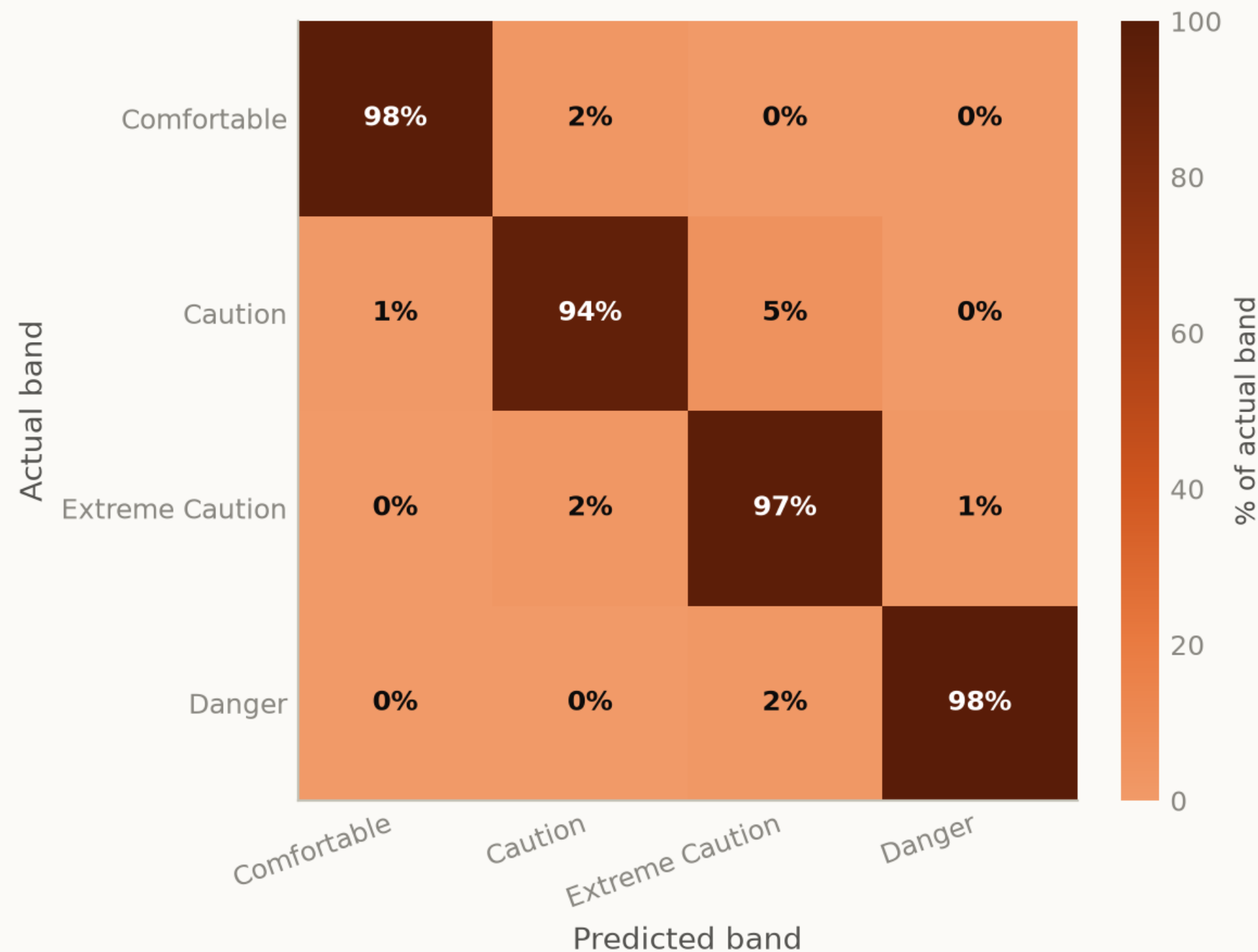
Distance to the crescent dominates at 0.85, 8.2× the next feature. Planting density still ranks above canopy width, so trees remain the cheaper lever — but on this layout the single strongest predictor of whether a square metre is shaded is simply how far it is from the crescent.  
Source: This project's models — see notebooks/ and src/ · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Fig07 confusion matrix

Analysis output — computed from project data.

Predicting thermal stress from the clock and the sun alone

Temperature and humidity withheld · test accuracy 97.5%



Rows sum to 100%. Errors fall almost entirely between adjacent bands, never between comfortable and dangerous.  
Source: This project's models — see notebooks/ and src/ · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge



# Every step of the work, with its proof attached

One row per phase: what was done, the code that did it, the file it wrote, and the picture that came out. Every file name is a live link — click it and read the actual thing, do not take this document's word for it.

## PHASE 9 AI Workflow & Visualisation

THIS DOCUMENT RESTS ON

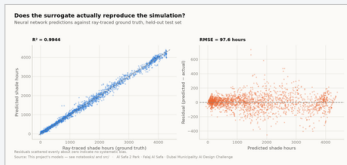


fig05\_surrogate\_performance.png

The four models and the anti-leakage discipline; drawings, boards, the analytics portal and the sixty-second film

CODE [src/models.py](#) [src/boards.py](#) [tools/sync\\_film.py](#)

PROOF — [click to open](#) [models/model\\_metrics.json](#) [tests/test\\_pipeline.py](#)

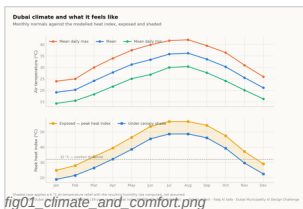


fig01\_climate\_and\_comfort.png

## PHASE 1 Site & Context Analysis

39 years of NCM normals rebuilt into an 8,760-hour year, verified back to within 0.39 °C; sun position for every hour via NREL/pvlib; shadow geometry; 7,640 residents inside a ten-minute walk

CODE [src/climate.py](#) [src/solar.py](#)

PROOF — [click to open](#) [data/processed/hourly\\_climate\\_comfort\\_8760.csv](#) [data/raw/sources.json](#)

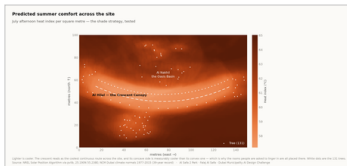


fig04\_site\_comfort\_map.png

## PHASE 2 Problem Definition

Phase 1 findings turned into problems and scored by severity rather than listed flat — thermal discomfort dominates everything else

CODE [src/climate.py](#)

PROOF — [click to open](#) [models/headline\\_metrics.json](#)

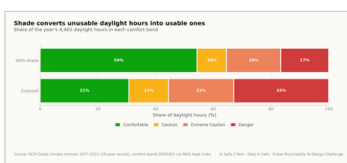


fig02\_comfort\_bands.png

## PHASE 3 Opportunity & Objectives

The ranked problems converted into measurable targets, including the comfort-hours target the finished design is scored against

CODE [src/config.py](#) [src/plan.py](#)

PROOF — [click to open](#) [models/headline\\_metrics.json](#)

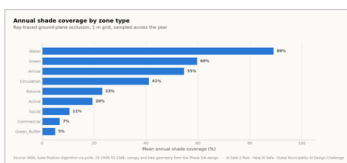


fig03\_shade\_by\_zone.png

## PHASE 4 Concept Development

Multiple plan forms generated and swept against the solar model; the crescent selected on hours-with-no-shade, not on mean coverage

CODE [src/plan.py](#) [src/solar.py](#)

PROOF — [click to open](#) [data/processed/masterplan\\_geometry.json](#)

# The work, with its proof attached — continued

One row per phase: what was done, the code that did it, the file it wrote, and the picture that came out. Every file name is a live link — click it and read the actual thing, do not take this document's word for it.



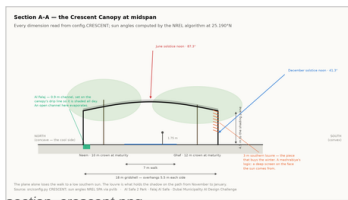
fig10\_masterplan.png

## PHASE 5 Masterplan Development

Every room struck off the crescent's centre; areas taken as the shoelace area of the drawn polygon, so the schedule closes on 15,000 m<sup>2</sup>

CODE [src/plan.py](#) [src/figures.py](#)

PROOF — click to open [data/raw/site\\_zoning\\_schedule.csv](#)  
[data/processed/masterplan\\_geometry.json](#)



section\_crescent.png

## PHASE 6 Detailed Design

The canopy section solved against shadow geometry — 7 m walk, 18 m gridshell at 4.5 m, 3 m southern louvre; 131 trees at mature canopy

CODE [src/drawings.py](#) [src/config.py](#)

PROOF — click to open [data/processed/planting\\_layout.csv](#)

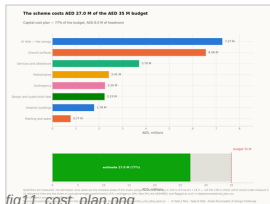


fig11\_cost\_plan.png

## PHASE 7 Performance & Sustainability

Water balance, carbon, the canopy PV deficit, and ray-traced shade performance by zone

CODE [src/costing.py](#) [src/climate.py](#)

PROOF — click to open [data/processed/cost\\_plan.csv](#) [models/cost\\_summary.json](#)

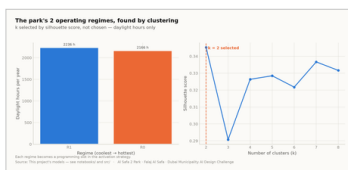


fig08\_microclimate\_regimes.png

## PHASE 8 User Experience & Activation

K-Means microclimate regimes and the hour-by-month comfort surface; programming targeted at late afternoon, spring and autumn

CODE [src/models.py](#) [src/dataset.py](#)

PROOF — click to open [data/processed/spatial\\_grid\\_comfort.csv](#)



board\_2\_evidence.png

## PHASE 10 Submission Assembly

Every report and visual mapped onto the twelve upload slots, each with a manifest naming what produced it

CODE [tools/build\\_submission\\_pdfs.py](#) [tools/build\\_reports.py](#)

PROOF — click to open [tests/test\\_pipeline.py](#)

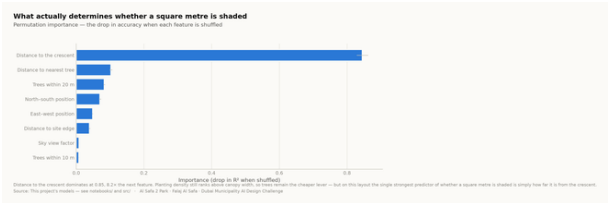
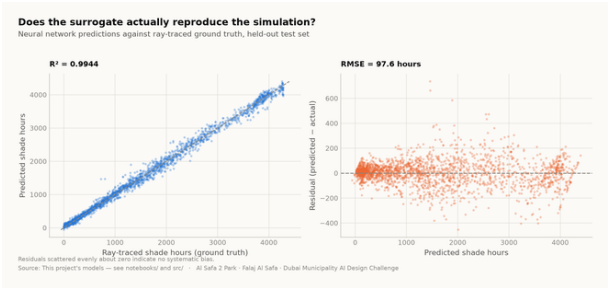
# The 3 images in this file, and the whole record

This slot's own pictures come first, larger. The remaining 15 of the 18 images the pipeline produces follow, so the whole visual record is in every file. Each links to its full-resolution original; nothing here is hand-drawn.

06

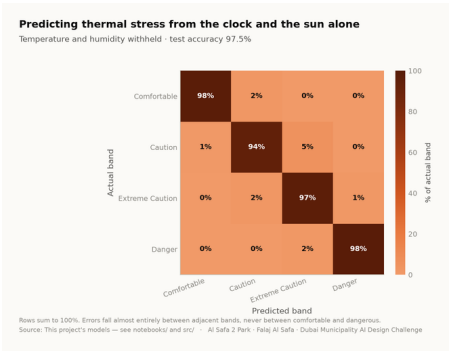
OF 12

## 01 IN THIS FILE



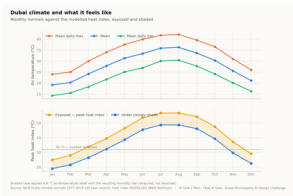
**Fig05 surrogate performance**  
analysis output

**Fig06 feature importance**  
analysis output

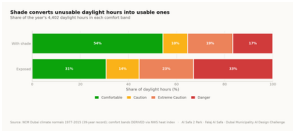


**Fig07 confusion matrix**  
analysis output

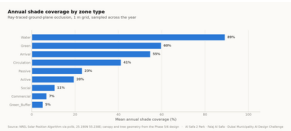
## 02 THE REST OF THE RECORD



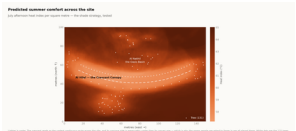
**Fig01 climate and comfort**  
analysis output



**Fig02 comfort bands**  
analysis output



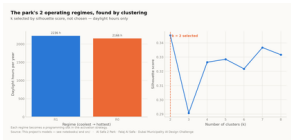
**Fig03 shade by zone**  
analysis output



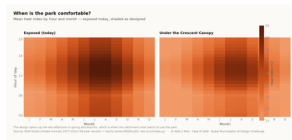
**Fig04 site comfort map**  
analysis output

# The visual record — continued

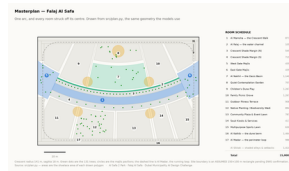
This slot's own pictures come first, larger. The remaining 15 of the 18 images the pipeline produces follow, so the whole visual record is in every file. Each links to its full-resolution original; nothing here is hand-drawn.



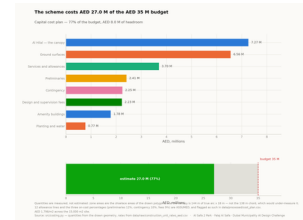
**Fig08 microclimate regimes**  
analysis output



**Fig09 diurnal comfort**  
analysis output



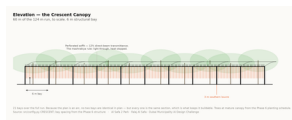
**Fig10 masterplan**  
analysis output



**Fig11 cost plan**  
analysis output



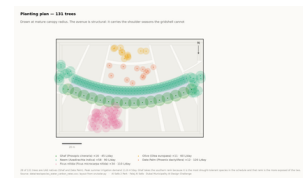
**Circulation**  
technical drawing



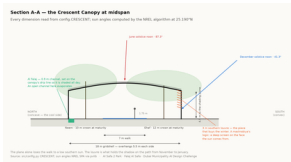
**Elevation**  
technical drawing



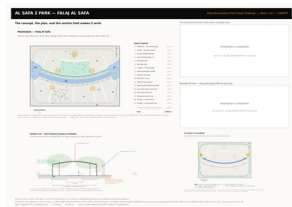
**Facilities**  
technical drawing



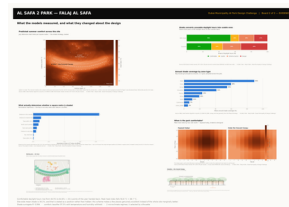
**Planting**  
technical drawing



**Section**  
technical drawing



**Board 1 concept**  
presentation board



**Board 2 evidence**  
presentation board

# Proof the analysis runs, and what it reports

These are the values the pipeline wrote on its last run, read straight out of the files it produced. The commands below regenerate every one of them from the published repository.

06

OF 12

## THE FIGURES THIS FILE LEADS ON

each one appears in the full tables below, from the same file

**R<sup>2</sup> 0.9944**

SHADE SURROGATE, TEST SET

**97.5%**

COMFORT BAND ACCURACY

**k = 2**

MICROCLIMATE REGIMES

**4,402**

DAYLIGHT HOURS MODELLED

### MEASURED RESULT

models/headline\_metrics.json

|                                          |         |
|------------------------------------------|---------|
| Annual daylight hours modelled           | 4,402   |
| Comfortable daylight hours — today       | 44.5%   |
| Comfortable daylight hours — as designed | 64.6%   |
| Mean heat-index reduction under canopy   | 7.13 °C |
| Peak heat index, exposed                 | 56.8 °C |
| Peak heat index, shaded                  | 48.7 °C |
| Crescent Walk shaded, canopy and louvre  | 87.3%   |
| Site-wide mean shade                     | 34.1%   |
| Trees planted                            | 131     |

### TRAINED MODELS, ON HELD-OUT TEST SETS

models/model\_metrics.json

|                                                              |                        |
|--------------------------------------------------------------|------------------------|
| M1a Random Forest — shade surrogate, test R <sup>2</sup>     | 0.9982                 |
| M1b Neural network — deployed surrogate, test R <sup>2</sup> | 0.9944                 |
| M2 Gradient Boosting — comfort band, test accuracy           | 97.5%                  |
| M3 K-Means — regimes selected by silhouette                  | k = 2                  |
| Training / validation / test split                           | 10,500 / 2,250 / 2,250 |
| Random seed, fixed for reproducibility                       | 42                     |

### WHAT THE CHECKS PROTECT

each one asserts what would otherwise drift silently

#### Climate fidelity

the modelled year reproduces the published normals it was built from

#### No overlaps

no room in the schedule overlaps another

#### Areas close

every area sums back to 15,000 m<sup>2</sup>

#### Budget held

the cost plan stays inside AED 35 M

#### No leakage

no model can see its own answer in its inputs

### RUN IT YOURSELF

```
git clone https://github.com/wasim437/Al_SAFA.git
pip install -r requirements.txt
python run_analysis.py           # data, models, figures
python -m tests.test_pipeline    # 41 correctness checks
node docs/_PORTAL/selftest.js   # 64 portal checks
node tests/test_film.js         # every frame of the film
```

# The complete project, and where to check it

Every one of the 121 links below is live. The repository holds the data as issued and as processed, the code, the trained models, the drawings and the tests, and it runs end to end with one command. A juror who wants to know where a number came from can be given a file path rather than an opinion.

Repository [github.com/wasim437/AI\\_SAFA](https://github.com/wasim437/AI_SAFA)  
Live portal [wasim437.github.io/AI\\_SAFA/](https://wasim437.github.io/AI_SAFA/)

## THE SOURCES BEHIND THIS FILE

*open these first if you want to check this document*

|                                           |                               |
|-------------------------------------------|-------------------------------|
| <a href="#">src/models.py</a>             | Technical drawing — to scale. |
| <a href="#">src/dataset.py</a>            | Technical drawing — to scale. |
| <a href="#">models/model_metrics.json</a> | Technical drawing — to scale. |
| <a href="#">tests/test_pipeline.py</a>    | Technical drawing — to scale. |

## THE TWELVE UPLOAD FILES (12)

|                                                                                              |                                              |
|----------------------------------------------------------------------------------------------|----------------------------------------------|
| <a href="#">UPLOAD_THESE_12_FILES/01_Design_Narrative_and_Concept.pdf</a>                    | Design Narrative & Concept                   |
| <a href="#">UPLOAD_THESE_12_FILES/02_Neighborhood_Park_Preliminary_Design_Masterplan.pdf</a> | Neighborhood Park Preliminary Design Ma...   |
| <a href="#">UPLOAD_THESE_12_FILES/03_Concept_Plans_and_Spatial_Organization_Diagrams.pdf</a> | Concept Plans and Spatial Organization Di... |
| <a href="#">UPLOAD_THESE_12_FILES/04_Key_Sections_and_Elevations.pdf</a>                     | Key Sections & Elevations                    |
| <a href="#">UPLOAD_THESE_12_FILES/05_3D_and_Spatial_Visualizations.pdf</a>                   | 3D & Spatial Visualizations                  |
| <a href="#">UPLOAD_THESE_12_FILES/06_AI_Methodology_Report.pdf</a>                           | AI Methodology Report                        |
| <a href="#">UPLOAD_THESE_12_FILES/07_User_Experience_and_Activation_Strategy.pdf</a>         | User Experience & Activation Strategy        |
| <a href="#">UPLOAD_THESE_12_FILES/08_Sustainability_Concept_and_Strategy.pdf</a>             | Sustainability Concept & Strategy            |
| <a href="#">UPLOAD_THESE_12_FILES/09_Material_and_Landscape_Palette.pdf</a>                  | Material & Landscape Palette                 |
| <a href="#">UPLOAD_THESE_12_FILES/10_Complete_Design_Report.pdf</a>                          | Complete Design Report                       |
| <a href="#">UPLOAD_THESE_12_FILES/11_Site_Analysis_and_Human-Centric_Research.pdf</a>        | Site Analysis & Human-Centric Research       |
| <a href="#">UPLOAD_THESE_12_FILES/12_One-minute_Concept_Animation.pdf</a>                    | One-minute Concept Animation                 |

## PROJECT DOCUMENTS (7)

|                                                   |                                                |
|---------------------------------------------------|------------------------------------------------|
| <a href="#">AL_SAFA_MASTER_PROMPT.md</a>          | The whole design, stated for visualisation     |
| <a href="#">DATA_SOURCES.md</a>                   | Every source, its period, and its limitations  |
| <a href="#">LINKS.md</a>                          | Every public URL this submission points at     |
| <a href="#">PROJECT_PLAN.md</a>                   | Requirements, phases, status, and what is left |
| <a href="#">PUSH_TO_GITHUB.md</a>                 | Project document                               |
| <a href="#">README.md</a>                         | The design argument, written for a juror       |
| <a href="#">EXPLAIN_THE_PROJECT/START_HERE.md</a> | The project in plain language                  |

## THE WRITTEN REPORTS (11)

|                                                                              |                                  |
|------------------------------------------------------------------------------|----------------------------------|
| <a href="#">reports/pdf/AI_Safa_2_Park_Complete_Design_Report.pdf</a>        | Generated from the live analysis |
| <a href="#">reports/pdf/Concept_Animation_Storyboard.pdf</a>                 | Generated from the live analysis |
| <a href="#">reports/pdf/Phase2_Problem_Definition_Report.pdf</a>             | Generated from the live analysis |
| <a href="#">reports/pdf/Phase3_Opportunity_and_Objectives_Report.pdf</a>     | Generated from the live analysis |
| <a href="#">reports/pdf/Phase4_Concept_Development_Report.pdf</a>            | Generated from the live analysis |
| <a href="#">reports/pdf/Phase5_Masterplan_Development_Report.pdf</a>         | Generated from the live analysis |
| <a href="#">reports/pdf/Phase6_Detailed_Design_Report.pdf</a>                | Generated from the live analysis |
| <a href="#">reports/pdf/Phase7_Performance_and_Sustainability_Report.pdf</a> | Generated from the live analysis |
| <a href="#">reports/pdf/Phase8_User_Experience_and_Activation_Report.pdf</a> | Generated from the live analysis |
| <a href="#">reports/pdf/Phase9_AI_Workflow_and_Visualization_Report.pdf</a>  | Generated from the live analysis |
| <a href="#">reports/pdf/Visualisation_Strategy_and_Image_Provenance.pdf</a>  | Generated from the live analysis |

Links resolve once the repository is published. GitHub renders PDFs, notebooks, CSVs and images in the browser — nothing needs to be downloaded or cloned.

# The complete project — continued

## THE ANALYSIS FIGURES (11)

[figures/fig01\\_climate\\_and\\_comfort.png](#)  
[figures/fig02\\_comfort\\_bands.png](#)  
[figures/fig03\\_shade\\_by\\_zone.png](#)  
[figures/fig04\\_site\\_comfort\\_map.png](#)  
[figures/fig05\\_surrogate\\_performance.png](#)  
[figures/fig06\\_feature\\_importance.png](#)  
[figures/fig07\\_confusion\\_matrix.png](#)  
[figures/fig08\\_microclimate\\_regimes.png](#)  
[figures/fig09\\_diurnal\\_comfort.png](#)  
[figures/fig10\\_masterplan.png](#)  
[figures/fig11\\_cost\\_plan.png](#)

Analysis output — computed from project data  
 Analysis output — computed from project data  
 Analysis output — computed from project data  
 Analysis output — computed from project data  
 Analysis output — computed from project data  
 Analysis output — computed from project data  
 Analysis output — computed from project data  
 Analysis output — computed from project data  
 Analysis output — computed from project data  
 Analysis output — computed from project data  
 Analysis output — computed from project data

## THE TECHNICAL DRAWINGS AND BOARDS (7)

[design/visuals/circulation\\_crescent.png](#)  
[design/visuals/elevation\\_crescent.png](#)  
[design/visuals/facilities\\_crescent.png](#)  
[design/visuals/planting\\_crescent.png](#)  
[design/visuals/section\\_crescent.png](#)  
[design/boards/board\\_1\\_concept.png](#)  
[design/boards/board\\_2\\_evidence.png](#)

Technical drawing — generated from `src/plan.py`  
 Technical drawing — generated from `src/plan.py`  
 Technical drawing — generated from `src/plan.py`  
 Technical drawing — generated from `src/plan.py`  
 Technical drawing — generated from `src/plan.py`  
 Technical drawing — generated from `src/plan.py`  
 Technical drawing — generated from `src/plan.py`

## THE ANALYSIS CODE (13)

[src/\\_\\_init\\_\\_.py](#)  
[src/boards.py](#)  
[src/climate.py](#)  
[src/config.py](#)  
[src/costing.py](#)  
[src/dataset.py](#)  
[src/drawings.py](#)  
[src/figures.py](#)  
[src/models.py](#)  
[src/plan.py](#)  
[src/solar.py](#)  
[src/viz.py](#)  
[run\\_analysis.py](#)

Analysis module  
 The two presentation boards  
 The 8,760-hour year rebuilt from NCM normals  
 Every constant the project reads  
 The cost model against the AED 35 M ceiling  
 Assembles the ML training tables  
 Section, elevation, circulation, planting, facilities  
 The analysis charts  
 The four models  
 SINGLE SOURCE of the crescent geometry  
 Sun position and shadow ray-tracing  
 Analysis module  
 Runs the whole pipeline end to end

## THE BUILD AND SYNC TOOLS (12)

[tools/\\_\\_init\\_\\_.py](#)  
[tools/build\\_docs.py](#)  
[tools/build\\_links.py](#)  
[tools/build\\_notebook.py](#)  
[tools/build\\_reports.py](#)  
[tools/build\\_submission\\_pdfs.py](#)  
[tools/embed\\_narration.py](#)  
[tools/organize\\_repo.py](#)  
[tools/report\\_content.py](#)  
[tools/sync\\_film.py](#)  
[tools/sync\\_portal.py](#)  
[tools/sync\\_submission.py](#)

# The complete project — continued

## THE DATA, AS ISSUED (7)

|                                                          |                                       |
|----------------------------------------------------------|---------------------------------------|
| <a href="#">data/raw/construction_unit_rates_aed.csv</a> | Source dataset                        |
| <a href="#">data/raw/dubai_climate_normals_ncm.csv</a>   | Source dataset                        |
| <a href="#">data/raw/dubai_population_dsc_2023.csv</a>   | Source dataset                        |
| <a href="#">data/raw/site_zoning_schedule.csv</a>        | The measured room schedule            |
| <a href="#">data/raw/sources.json</a>                    | Every source dataset, with its period |
| <a href="#">data/raw/species_water_carbon_rates.csv</a>  | Source dataset                        |
| <a href="#">data/raw/walk_catchment_rings.csv</a>        | Source dataset                        |

## THE DATA, AS PROCESSED (6)

|                                                                |                                           |
|----------------------------------------------------------------|-------------------------------------------|
| <a href="#">data/processed/cost_plan.csv</a>                   | The capital cost plan, line by line       |
| <a href="#">data/processed/hourly_climate_comfort_8760.csv</a> | The 8,760-hour climate and comfort series |
| <a href="#">data/processed/masterplan_geometry.json</a>        | The plan geometry, as exported            |
| <a href="#">data/processed/planting_layout.csv</a>             | Every one of the 131 trees                |
| <a href="#">data/processed/schema.json</a>                     | Generated by the pipeline                 |
| <a href="#">data/processed/spatial_grid_comfort.csv</a>        | The 15,000-cell ground grid               |

## THE TRAINED MODELS (3)

|                                              |                       |
|----------------------------------------------|-----------------------|
| <a href="#">models/cost_summary.json</a>     | Model output          |
| <a href="#">models/headline_metrics.json</a> | The headline numbers  |
| <a href="#">models/model_metrics.json</a>    | Trained-model metrics |

## THE TESTS (2)

|                                        |                                 |
|----------------------------------------|---------------------------------|
| <a href="#">tests/test_pipeline.py</a> | 41 correctness checks           |
| <a href="#">tests/test_film.js</a>     | Every frame of the concept film |

## THE NOTEBOOK (1)

|                                                                  |                                         |
|------------------------------------------------------------------|-----------------------------------------|
| <a href="#">notebooks/AL_SAFA_2_PARK_COMPLETE_ANALYSIS.ipynb</a> | The complete analysis, outputs embedded |
|------------------------------------------------------------------|-----------------------------------------|

## THE WEBSITE AND ANALYTICS PORTAL (9)

|                                               |                     |
|-----------------------------------------------|---------------------|
| <a href="#">docs/index.html</a>               | The project website |
| <a href="#">docs/SUBMISSION_GUIDE.md</a>      | Published site      |
| <a href="#">docs/_PORTAL/gallery_data.js</a>  | Published site      |
| <a href="#">docs/_PORTAL/plan_geometry.js</a> | Published site      |
| <a href="#">docs/_PORTAL/portal.js</a>        | Published site      |
| <a href="#">docs/_PORTAL/portal_data.js</a>   | Published site      |
| <a href="#">docs/_PORTAL/selftest.js</a>      | Published site      |
| <a href="#">docs/_PORTAL/DATA_AUDIT.md</a>    | Published site      |
| <a href="#">docs/_PORTAL/README.md</a>        | Published site      |

## THE COMPETITION BRIEF, AS ISSUED (7)

|                                                                                             |                                    |
|---------------------------------------------------------------------------------------------|------------------------------------|
| <a href="#">00_BRIEF/Ai Park - Master Plan (4).jpg</a>                                      | Dubai Municipality source document |
| <a href="#">00_BRIEF/Ai Park Scope of Work &amp; General Terms &amp; Conditions (5).pdf</a> | Dubai Municipality source document |
| <a href="#">00_BRIEF/AI Safa Park 2 Plan (5).dwg</a>                                        | Dubai Municipality source document |
| <a href="#">00_BRIEF/MAIN WEBSITE DETAILS.txt</a>                                           | Dubai Municipality source document |
| <a href="#">00_BRIEF/Neighborhood Parks Manual (4).pdf</a>                                  | Dubai Municipality source document |
| <a href="#">00_BRIEF/NEIGHBORHOOD PARKS MANUAL_TEXT.txt</a>                                 | Dubai Municipality source document |
| <a href="#">00_BRIEF/SCOPE_OF_WORK_FULL_TEXT.txt</a>                                        | Dubai Municipality source document |

## THE TWELVE SUBMISSION FOLDERS (12)

|                                                                          |                                                  |
|--------------------------------------------------------------------------|--------------------------------------------------|
| <a href="#">submission/01_Design_Narrative_Concept/MANIFEST.md</a>       | What is in the slot, and what produced each file |
| <a href="#">submission/02_Preliminary_Design_Masterplan/MANIFEST.md</a>  | What is in the slot, and what produced each file |
| <a href="#">submission/03_Concept_Plans_Spatial_Diagrams/MANIFEST.md</a> | What is in the slot, and what produced each file |
| <a href="#">submission/04_Key_Sections_Elevations/MANIFEST.md</a>        | What is in the slot, and what produced each file |

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# The complete project — continued

[submission/05\\_3D\\_Spatial\\_Visualizations/MANIFEST.md](#)  
[submission/06\\_AI\\_Methodology\\_Report/MANIFEST.md](#)  
[submission/07\\_User\\_Experience\\_Activation\\_Strategy/MANIFEST.md](#)  
[submission/08\\_Sustainability\\_Concept\\_Strategy/MANIFEST.md](#)  
[submission/09\\_Material\\_Landscape\\_Palette/MANIFEST.md](#)  
[submission/10\\_Complete\\_Design\\_Report/MANIFEST.md](#)  
[submission/11\\_Site\\_Analysis\\_Human\\_Centric\\_Research/MANIFEST.md](#)  
[submission/12\\_Concept\\_Animation\\_Video/MANIFEST.md](#)

What is in the slot, and what produced each file  
What is in the slot, and what produced each file  
What is in the slot, and what produced each file  
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What is in the slot, and what produced each file  
What is in the slot, and what produced each file  
What is in the slot, and what produced each file  
What is in the slot, and what produced each file

WORKING HISTORY (1)

[archive/withdrawn\\_visuals/README.md](#)

Images withdrawn on purpose, and why